Day 1 Documentation

IoT — Day 1 Starter Task
Topic: Getting familiar with ESP32-S3

Today, focus on understanding the ESP32-S3 before working deeply with the sensors.

Do these 3 things:

1. Learn the basic architecture/features of the ESP32-S3.
2. Set up your ESP32 development environment — PlatformIO or ESP-IDF.
3. Create and run a simple Hello World / Serial output program.

Also study briefly:

- ESP32-S3
- Embedded C/C++
- GPS/GNSS
- IMU
- Accelerometer
- Gyroscope

You don't need to master everything today.

By the end of the session, you should be able to explain:

- What the ESP32-S3 does
- What GPS/GNSS provides
- What an IMU provides
- Difference between accelerometer and gyroscope

If you have the hardware, run the serial program on the board. If not, use the assigned simulator and document what you tested.

Choose one GPS/GNSS sensor/module and one IMU sensor that could realistically be used in a vehicle-tracking/safety prototype.

For each one, give:

- Part/module name
- What it measures
- Interface
- Supply voltage
- Why it is suitable
- One limitation
- Approximate practical cost range

Keep it to one page/table.

ESP32-S3 :

The ESP32-S3 is a 32-bit dual-core microcontroller from Espressif. It provides Wi-Fi and Bluetooth Low Energy along with GPIO, ADC, UART, I²C, SPI and USB interfaces. It can be used to collect sensor data, process it and communicate the information to other systems.

Key features :

Feature	Description
CPU	**Dual-core Xtensa LX7**
Maximum frequency	**Up to 240 MHz**
Wi-Fi	**2.4 GHz**
Bluetooth	**Bluetooth Low Energy**
GPIO	**Programmable digital I/O**
Communication	**UART, I²C, SPI**
USB	**USB OTG**
ADC	**Analog-to-digital conversion**
Application	**IoT, embedded systems, sensor processing**

GPS/GNSS:

- GPS = Global Positioning System → operated by the USA.
- GNSS = Global Navigation Satellite System

A GNSS receiver can determine things such as:

- Latitude
- Longitude
- Altitude
- Speed
- Time
- Direction/course

IMU Inertial Measurement Unit

It contains:

- Accelerometer
- Gyroscope

Accelerometer → Linear acceleration ,example : Sudden braking

It can help detect:

- Sudden acceleration
- Sudden braking
- Impact/crash
- Vehicle vibration
- Tilt

Gyroscope → Angular velocity/rotation ,example : Vehicle turning/tilting

Measures **angular velocity**, usually in °/s or rad/s.

It can detect:

- Rotation
- Turning
- Angular movement
- Vehicle orientation changes

GPS/GNSS → NEO-M8N

IMU → MPU6050

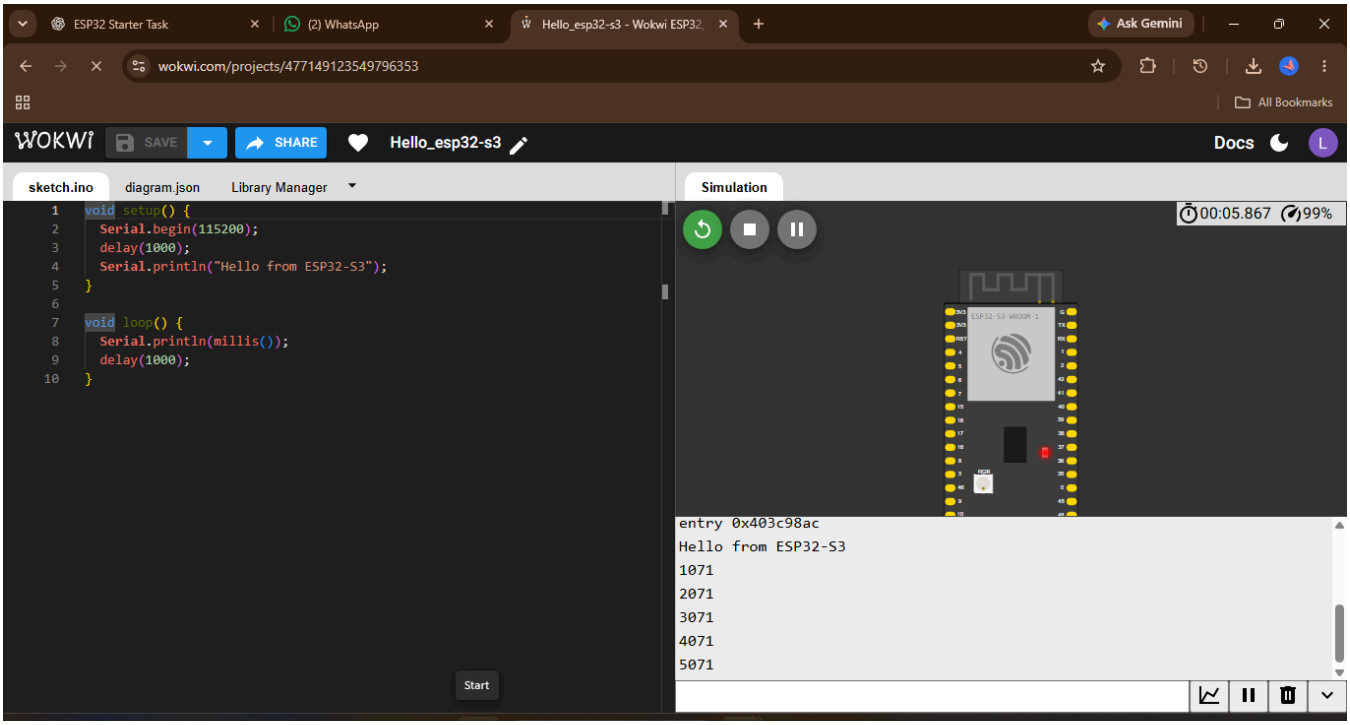
Parameter	GPS/GNSS	IMU
Part/module	**u-blox NEO-M8N GNSS module**	**MPU6050 6-axis IMU module**
What it measures	Position, speed, altitude, time, course	3-axis acceleration + 3-axis angular velocity
Interface	UART (commonly)	I²C
Supply voltage	Module typically accepts ~3.3–5 V depending on breakout board; GNSS IC itself is 3.3 V class	Breakout modules commonly accept ~3.3–5 V; sensor itself is 3.3 V
Why suitable	Reliable positioning for vehicle tracking; supports multiple GNSS constellations	Useful for detecting acceleration, braking, impacts, vibration and vehicle rotation
One limitation	Needs satellite visibility; indoor/urban environments can reduce accuracy or delay position fix	Sensor drift/noise means it cannot provide reliable absolute position by itself
Approx. practical cost in India	~₹1,500–₹3,000 for a typical NEO-M8N breakout/module	~₹150–₹400 for a typical MPU6050 breakout

Embedded C :

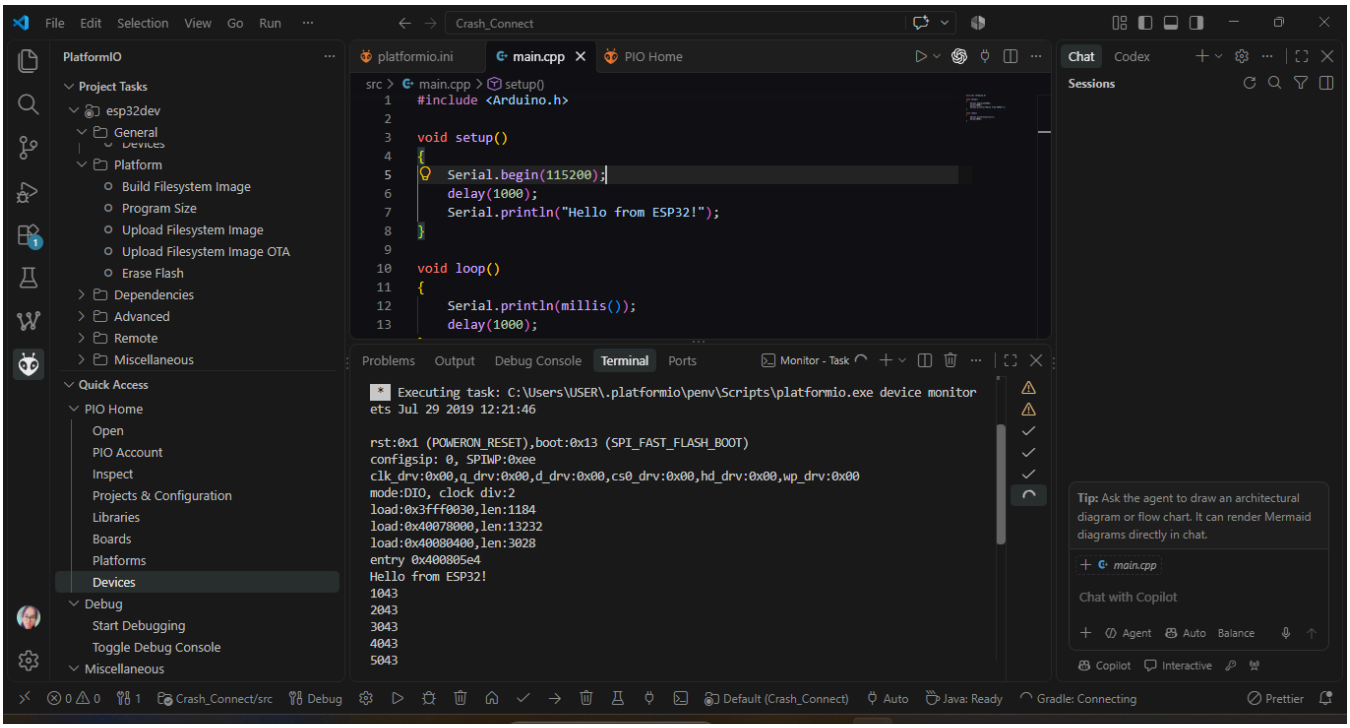
```
void setup()
{
   Serial.begin(115200);
}

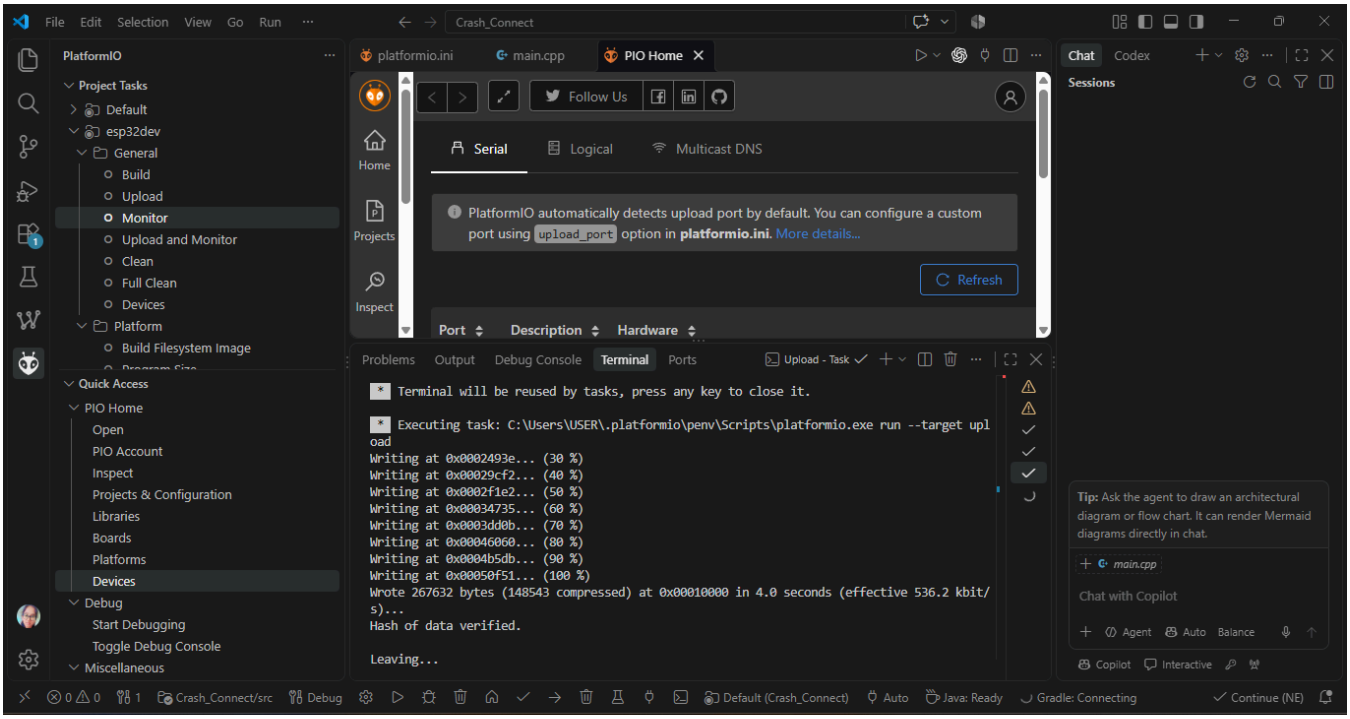
void loop()
{
   Serial.println("Hello World from ESP32-S3");
   delay(1000);
}
```

Wokwi Simulation :



PlatformIO SImulation :


File Edit Selection View Go Run
Crash_Connect
PlatformIO
Project Tasks
esp32dev
General
Platform
Build Filesystem Image
Program Size
Upload Filesystem Image
Upload Filesystem Image OTA
Erase Flash
Dependencies
Advanced
Remote
Miscellaneous
Quick Access
PIO Home
Open
PIO Account
Inspect
Projects & Configuration
Libraries
Boards
Platforms
Devices
Debug
Start Debugging
Toggle Debug Console
Miscellaneous
platformio.ini
main.cpp
PIO Home
src > main.cpp > setup()
#include <Arduino.h>
void setup()
{
Serial.begin(115200);
delay(1000);
Serial.println("Hello from ESP32!");
}
void loop()
{
Serial.println(millis());
delay(1000);
Problems Output Debug Console Terminal Ports
Monitor - Task
Executing task: C:\Users\USER\.platformio\penv\Scripts\platformio.exe device monitor
ets Jul 29 2019 12:21:46
rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4
Hello from ESP32!
1043
2043
3043
4043
5043
Chat
Codex
Sessions
Tip: Ask the agent to draw an architectural diagram or flow chart. It can render Mermaid diagrams directly in chat.
main.cpp
Chat with Copilot
Agent
Auto
Balance
Copilot
Interactive
Crash_Connect/src
Debug
Default (Crash_Connect)
Auto
Java: Ready
Gradle: Connecting
Prettier


File Edit Selection View Go Run
Crash_Connect
PlatformIO
Project Tasks
Default
esp32dev
General
Build
Upload
Monitor
Upload and Monitor
Clean
Full Clean
Devices
Platform
Build Filesystem Image
Quick Access
PIO Home
Open
PIO Account
Inspect
Projects & Configuration
Libraries
Boards
Platforms
Devices
Debug
Start Debugging
Toggle Debug Console
Miscellaneous
platformio.ini
main.cpp
PIO Home
Follow Us
Home
Projects
Inspect
Serial
Logical
Multicast DNS
PlatformIO automatically detects upload port by default. You can configure a custom port using upload_port option in platformio.ini. More details...
Refresh
Port
Description
Hardware
Problems Output Debug Console Terminal Ports
Upload - Task
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\USER\.platformio\penv\Scripts\platformio.exe run --target upl
oad
Writing at 0x0002493e... (30 %)
Writing at 0x00029cf2... (40 %)
Writing at 0x0002f1e2... (50 %)
Writing at 0x00034735... (60 %)
Writing at 0x0003dd0b... (70 %)
Writing at 0x00046060... (80 %)
Writing at 0x0004b5db... (90 %)
Writing at 0x00050f51... (100 %)
Wrote 267632 bytes (148543 compressed) at 0x00010000 in 4.0 seconds (effective 536.2 kbit/
s)...
Hash of data verified.
Leaving...
Chat
Codex
Sessions
Tip: Ask the agent to draw an architectural diagram or flow chart. It can render Mermaid diagrams directly in chat.
main.cpp
Chat with Copilot
Agent
Auto
Balance
Copilot
Interactive
Crash_Connect/src
Debug
Default (Crash_Connect)
Auto
Java: Ready
Gradle: Connecting
Continue (NE)